HARP: Human-Aware Routing Policies for Multi-Stage Decision Pipelines

Rohit Kulkarni, Lagnajeet Panigrahi
University of California, Berkeley

***Abstract*—A concise summary of the motivation, approach, and main findings of the work. Avoid citations and equations. Typically 150–250 words.**

***Keywords*—** keyword1, keyword2, keyword3, keyword4

I. Introduction

High-stakes institutional decision like medical insurance claims, immigration applications, and credit approvals are increasingly supported by machine learning systems. In general, these systems are not fully automated decision makers. Instead, automated models are embedded inside structured human review pipelines, where cases can be autonomously resolved or escalated to one or more human reviewers depending on factors including uncertainty, risk, and institutional constraints [1], [2].

This problem is called *selective prediction*, where a model is allowed to abstain on inputs where it has low confidence and defer to an expert human decision-maker [3], [4]. Previous works have studied selective classification and abstention, describing tradeoffs between prediction risk and coverage. Works such as [4], [5] have developed confidence-based deferral policies with strong theoretical guarantees. These approaches have been shown to be effective in applications where the alternative to automated prediction is a single, homogenous human expert.

However, real-world decision pipelines rarely conform to single-stage abstraction. In practice, human review is heirarchical, where casis can be routed to reviewers with differing levels of expertise, authority, and cost. For example, in insurance claim resolution, an application may be resolved automatically, reviewed by a junior examiner, escalated to a senior specialist, or forwarded to an administrative judge. Modelling deferral as a binary decision process does not always model this system well as it obscures important tradeoffs between accuracy and human resource allocation.

Previous work on algorithmic triage and human-AI collaboration has studied how automated systems can allocate cases between humans and models [6], [7], [8]. While these approaches acknowledge heterogenous human expertise, they focus on two-stage settings, rarely on heuristic or fixed routing rules, or do not explicitly connect to the selective prediction framework. As a result, existing methods provide limited guidance for designing multi-stage pipelines under explicit budget constraints.

In this work, we introduce Human-Aware Routing Policies (HARP), a framework for selecive prediction in multi-stage human-AI decision pipelines. Rather than making deferral a binary action, HARP formulates decision making as a routing problem over a heirarchy of human reviewers with increasing costs, with the objective of minizing system loss while also respective human resource consstraints.

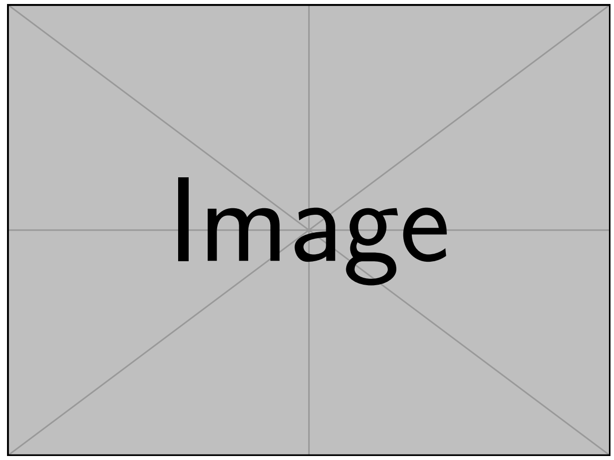

Fig. 1. Example figure caption.

We study this framework in the context of structured application review pipelines motivated by Social Security Disability Insurance (SSDI) claim resolution, where decisions are sequential, resource constrained, and subject to high institutional stakes. This setting shows the limitations of single-stage selective prediction and motivats the need for routing-aware decision policies.

Our work makes the following contributions:

- contrib1
- contrib 2
- contrib 3

II. Problem Formulation

A. Single Stage Policies

We specifically study human-AI collaborative decision making in structured application review pipelines, where an automated system can either issue a decision or defer the decision to a human reviewer at some cost.

Let

$(x, y) \sim D$, where D is the real world distribution over applications and ground-truth decisions

$x \in \mathcal{X}$, where x is an application

$y \in \mathcal{Y}$, where y is the ground truth decision

We consider a predictive model $f : \mathcal{X} \to \mathcal{Y}$ and a deferral policy $r : \mathcal{X} \to \{0, 1\}$, where

$r(x) = 0$ when the policy chooses to predict autonomously

$r(x) = 1$ when the policy defers to a human for a cost $c_h > 0$.

When $r(x) = 0$, the automated system incurs cost $\mathcal{L}_1$, which is given by

$$\mathcal{L}_1(f) = \ell(f(x), y),$$

where ℓ is by convention 0-1 loss or cross-entropy.

When $r(x) = 1$, the system defers to a human for a human cost $c_h > 0$, assuming that the human produces the correct decision. Therefore, the total system loss can be modeled as the expectation of a function of f and r:

$$\mathcal{L}(f, r) = \mathbb{E}_{(x,y)\sim D}\Big[\ell(f(x), y) \cdot \mathbf{1}\{r(x) = 0\} + c_h \cdot \mathbf{1}\{r(x) = 1\}\Big] \quad (1)$$

In this selective prediction model, when the policy $r(x) = 0$, f predicts autonomously. Over time, as $r(x)$ continues to be 1, c_h will increase. Therefore, we can define a coverage function of r as follows:

$$\operatorname{cov}(r) = \mathbb{E}_{(x,y)\sim D}[\mathbf{1}\{r(x) = 0\}], \quad (2)$$

which describes the fraction for which the policy r chooses for f not to abstain.

When $r(x) = 0$, $f(x)$ is evaluated and we can define the conditional risk as the loss ℓ between $f(x)$ and y for a given $(x, y) \sim D$. More formally, this is:

$$\operatorname{risk}(f, r) = \mathbb{E}_{(x,y)\sim D}[\ell(f(x), y) \mid r(x) = 0], \quad (3)$$

which shows the conditional risk when f autonomously makes the decision.

Based on this coverage function and the conditional risk $\mathcal{L}$, we can define a coverage-constrained optimization problem as follows:

$$\min_{f,r} \mathbb{E}\big[\ell(f(x), y) \mid r(x) = 0\big], \qquad \text{s.t. } \operatorname{cov}(r) \geq \kappa, \quad (4)$$

where κ is the constraint for the fraction where $r(x) = 0$.

Therefore, we can apply a confidence-based threshold defined on the confidence from the output of f, which is traditionally just the max of the output of the sigmoid or softmax function if f is a neural network. Thus, we let:

$p_\theta(y|x)$ be the predictive distribution of f (5)

$\operatorname{conf}(x) = \max_y p_\theta(y|x)$ be defined as the confidence of a given application-decision pair. (6)

We can then threshold the policy r on the confidence function for a hyperparameter τ as follows:

$$r_\tau(x) = \begin{cases} 0 & \text{if } \operatorname{conf}(x) \geq \tau, \\ 1 & \text{if } \operatorname{conf}(x) < \tau \end{cases} \quad (7)$$

B. Extension to Multi-Stage Policies

The optimization problem in Eq. (4) is applicable when we have a binary decision of using policy f or abstaining and deferring to an arbitrary human reviewer. However, when we have a heirarchical set of k human reviewers $H = \{0, 1, \ldots, k\}$, each with increasing costs c_{h_k}, we have c_{h_k} increasing when $k \geq 1$.

For the purposes of this paper, single-stage refers to $|H| = 1$, and multi-stage is when $|H| > 1$. We extend the single stage-policy to multi-stage. In this case, we make a trivial change to the minimization problem in Eq. (4) and the initial routing policy r. Let

$r_m : \mathcal{X} \to \{0, 1, \ldots, k\}$ be the multistage routing policy, (8)

$\ell_s(x)$ be the loss of sending x to stage s, (9)

B be the total budget constraint for the human resources available. (10)

We assume $\ell_0(x) = \ell(f(x), y)$ corresponds to the automated model loss, and $\ell_s(x) = c_{h_s}$ for $s \geq 1$, assuming human reviewers produce correct decisions.

Therefore, the constrained optimization problem we must solve is as follows:

$$\min_{r_m} \mathbb{E}\big[\ell_{r_m(x)}(x)\big], \qquad \text{s.t. } \mathbb{E}\big[c_{h_{r_m(x)}}\big] \leq B. \quad (11)$$

C. Stochastic Routing Policies

In practice, stochastic routing allows for calculating gradients in optimization, where the system learns a probabilistic policy over decision stages. Rather than deterministically assigning an application x to a single stage, we define a routing distribution $\pi_\phi(s \mid x)$ over stages $s \in \{0, 1, \ldots, k\}$, parameterized by ϕ. For each input, a stage is sampled according to this distribution, allowing the routing policy to express uncertainty and to smooth decisions near stage boundaries.

Under this formulation, the expected system loss is given by

$$\mathbb{E}_{(x,y)\sim D}\left[\sum_{s=0}^{k} \pi_\phi(s \mid x)\, \ell_s(x)\right], \quad (12)$$

with an associated expected human cost

$$\mathbb{E}_{x\sim D}\left[\sum_{s=0}^{k} \pi_\phi(s \mid x)\, c_{h_s}\right]. \quad (13)$$

Therefore, at training time the system is optimized stochastically, but during inference it is deterministic.

III. Background and Related Work

We overview selective clasification and prediction policies, algorithmic triage, and organizational resource management models in current literature.

A. Selective Prediction and The Reject Option

The basis for HARP is selective prediction, where a classifier has a reject option to abstain from making a prediction when a defined uncertainty score is high [3]. These works generally deal with the statistical modelling of the trade-off bewtween risk, which is the error rate, and coverage, the fraction of samples classified by the autonomous model. While previous literature has established confidence-based thresholds with statistical guarantees [4], they typically treat abstention as the end-state of the system. However, in institutional pipelines abstaining is not always the end of the decision, with several heirarchies of human reviewers analyzing samples for different costs.

B. Calibrated Selective Classification (CSC) as a Baseline

One of the main challenges in the model routing domain is ensuring that the error probability of a routing model is accurately captured by model confidence scores. Work by [5], [9] in Calibrated Selective Classification (CSC) has developed methods to provide statistical guarantees on risk by calibrating confidence scores within a separate, non-routing model.

While existing works in CSC typically focus on the binary "predict" or "defer" case, we extend these statistical guarantees to serve as a multi-stage baseline. Specifically, by calibrating a sequence of thresholds to partition a confidence space into n distinct regions, where n represents the number of reviewers in the institutional heirarchy, we create a strong calibration-based benchmark to compare with HARP's learned routing policies.

C. Learning to Defer (L2D)

Learning to Defer, or L2D, is a deferral policy where the automated decision-making model and the deferral policy should be optimized together to account for when there exists a human partner [10]. L2D generally assumes a single-stage system with a homogenous pool of experts [7] or just one expert. Our work extends L2D by recognizing that human expertise can be structured heirarchically, especially when cases are escalated to increasingly accurate human reviewer and higher costs are incurred on the institution.

IV. Data Processing

A. Data Source

B. Synthetic Data Generation

V. Model Architecture

VI. Results

Present results objectively, using tables and figures where appropriate.

VII. Discussion

Interpret the results, explain trends, limitations, and unexpected outcomes. Relate findings back to the motivation and prior work.

VIII. Conclusion and Future Work

Summarize key contributions and outline potential directions for future research.

Author Note

The authors declare no competing interests.

Author Contributions

Describe each author's contribution to the work.

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